

ECE264D: CMOS Analog Integrated Circuits and Systems IV

Lecture 14: Sub-sampling PLL and Sampling PLL

Yiwu Tang

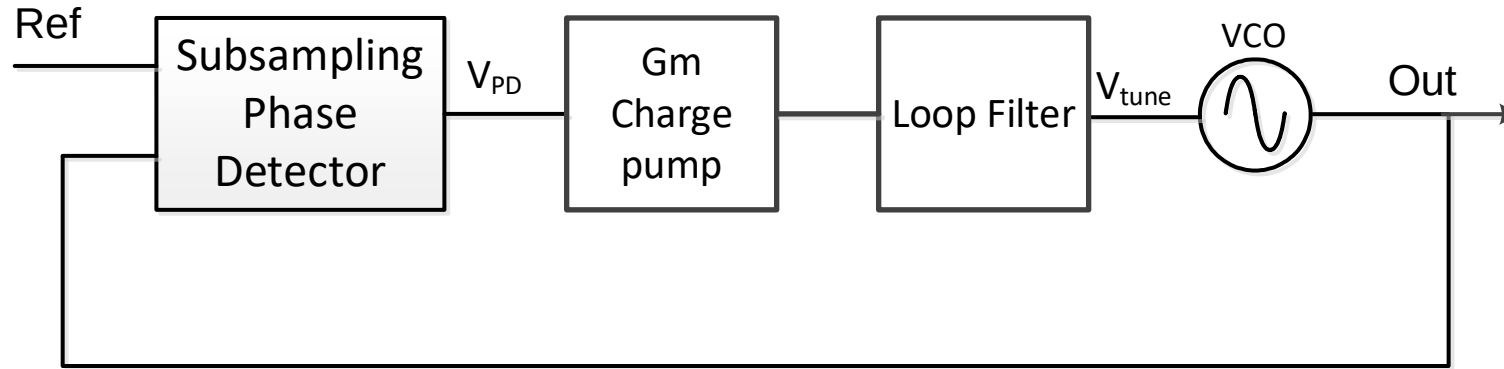


Outline

- Subsampling PLL architecture
 - Subsampling phase detector
 - Gm charge pump
 - Auxiliary frequency lock loop
 - Other building blocks
- Sampling PLL architecture
 - Sampling phase detector
- Subsampling and sampling PLL s-domain models
- Fractional-N operation



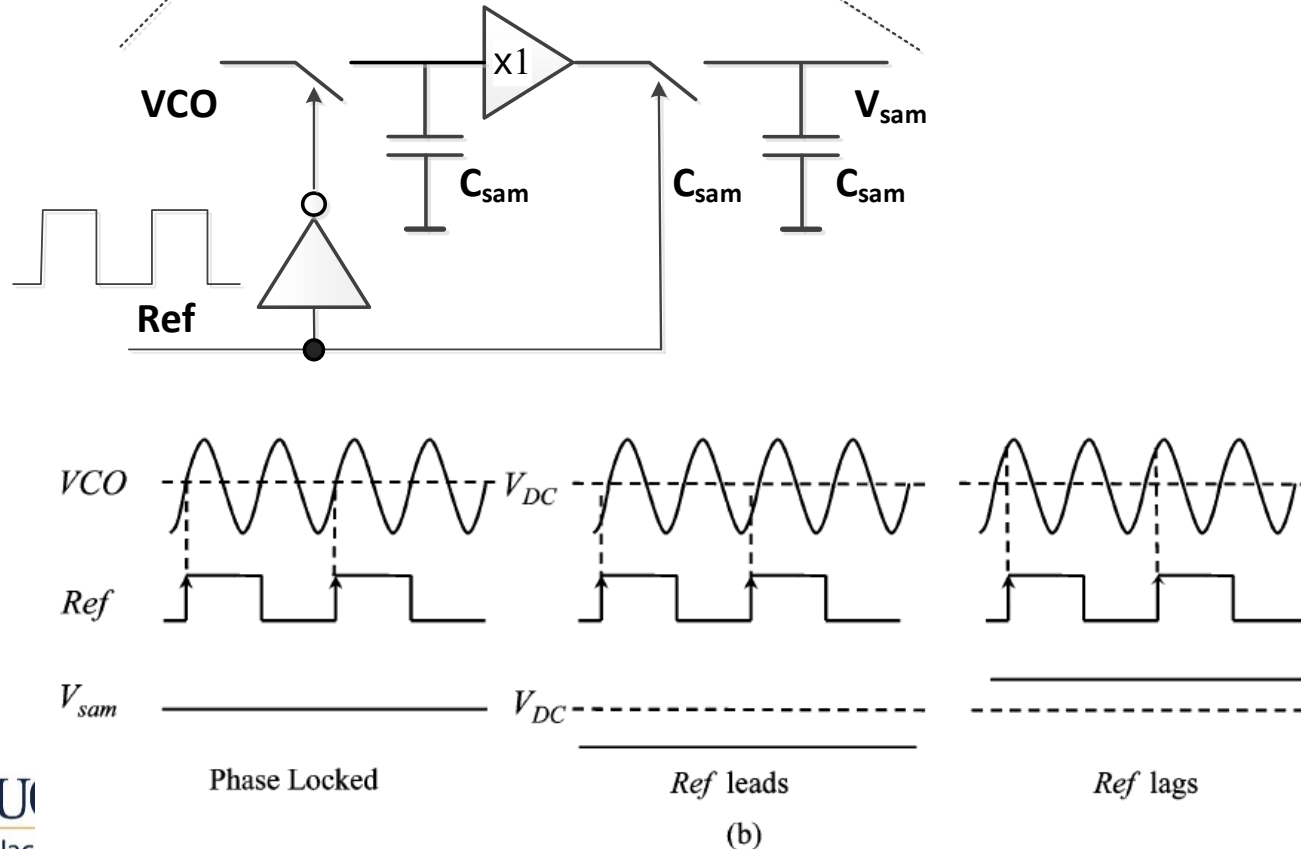
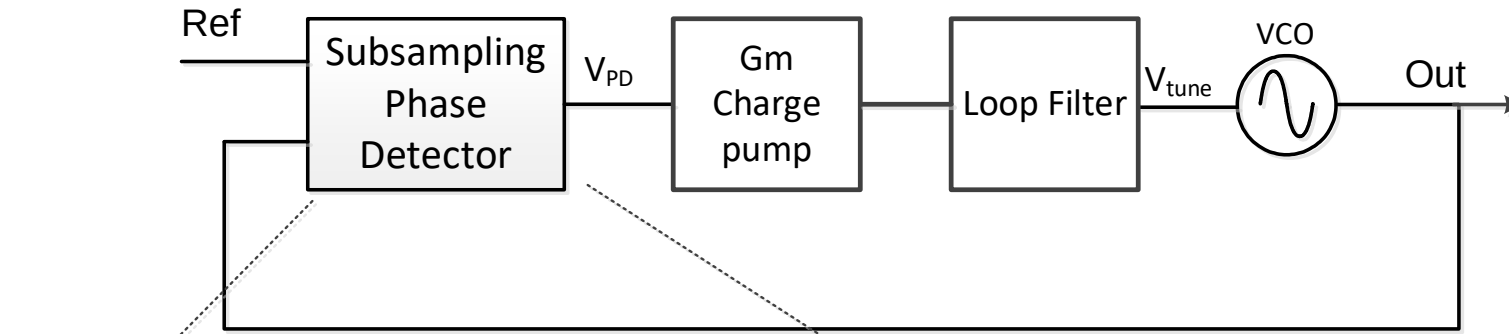
Subsampling PLL Architecture



- Reference clock directly samples VCO signal, no N-divider in the feedback path
- Sampled voltage is converted to current pulses that charge or discharge the loop filter
- N ambiguity – need an auxiliary frequency lock loop
- Integer-N PLL only – need DTC for fractional-N functionality



Subsampling PLL – Sampler

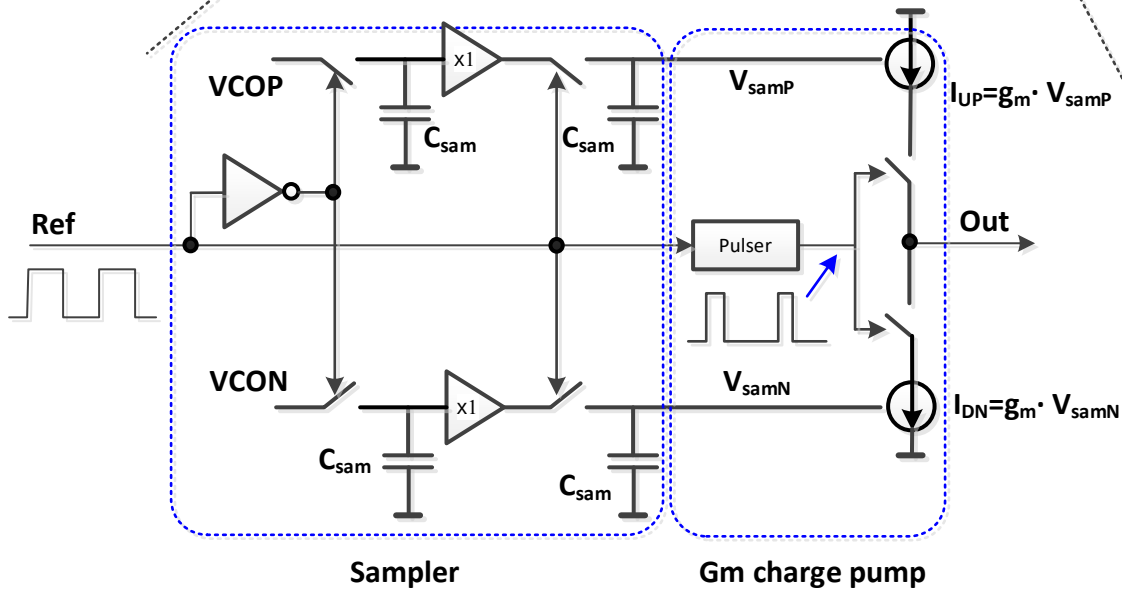
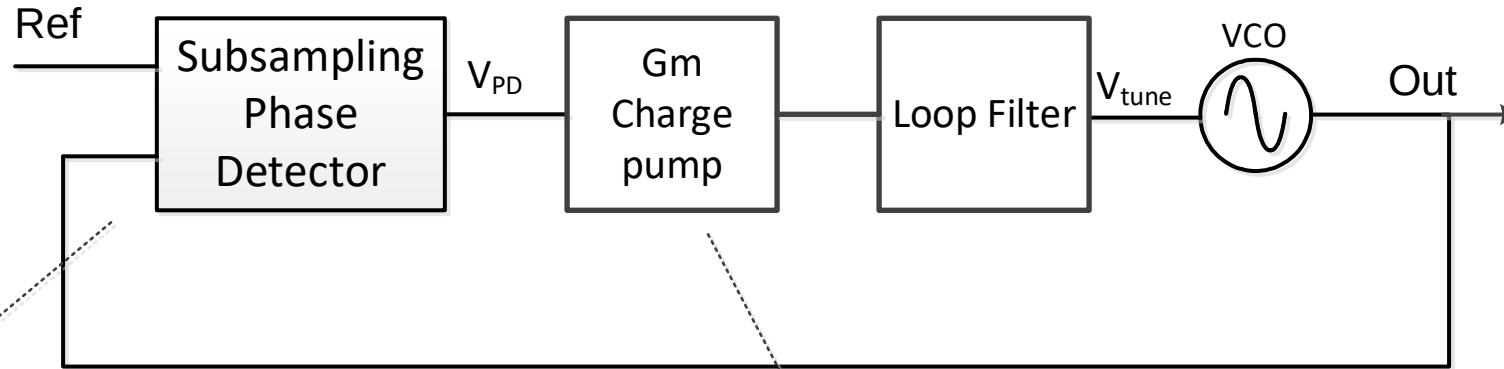


- When $F_{VCO} \neq N \cdot F_{ref}$, sampled voltage varies with the beat frequency
- When $F_{VCO} = N \cdot F_{ref}$
 - If Ref leads VCO, $V_{sam} < V_{DC}$
 - If Ref lags VCO, $V_{sam} > V_{DC}$
 - If Ref aligns with VCO, $V_{sam} = V_{DC}$
- V_{sam} carries phase difference information
- Is this type-I or type-II PLL?
- Does PLL lock at $V_{sam} = V_{DC}$?

Ref [1]



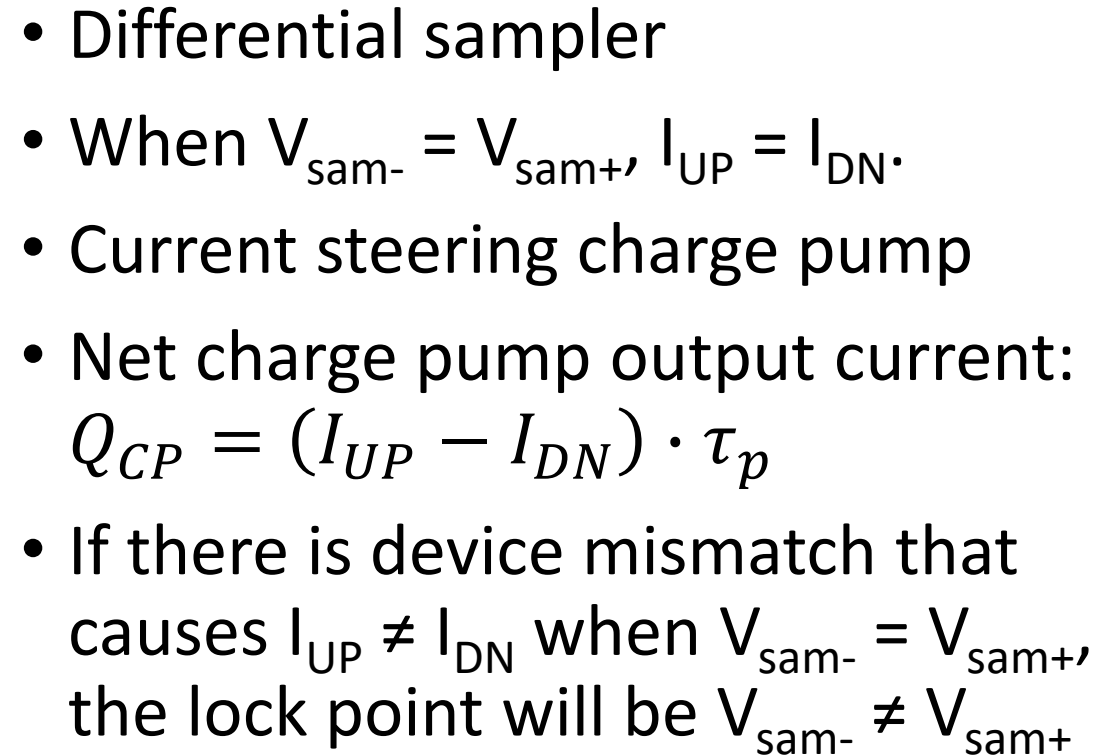
Subsampling PLL – Gm Charge Pump



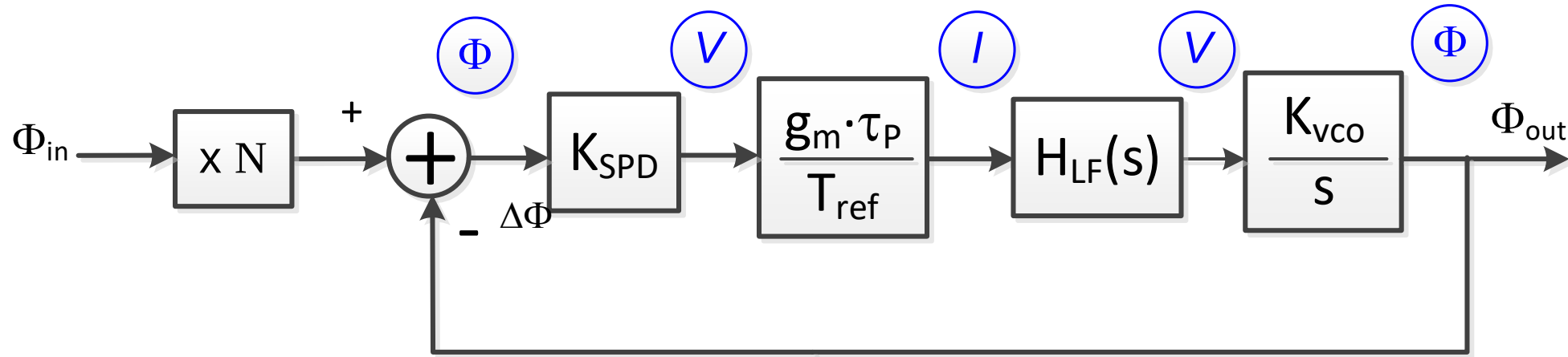
- Differential sampling, lock at $VCOP/N$ differential zero crossing
- Charge pump current is a function of the sampled voltage, G_m controls charge pump gain
- Pulser generates narrow pulses to
 - Reduce sampler gain
 - Saves power
 - Reduces noise
- Phase information is in the sampled voltage, not pulse width



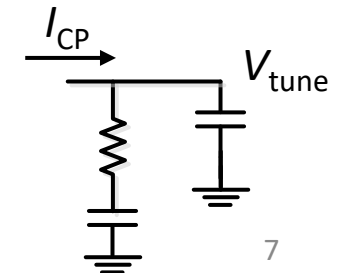
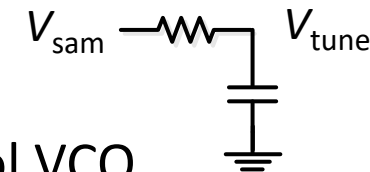
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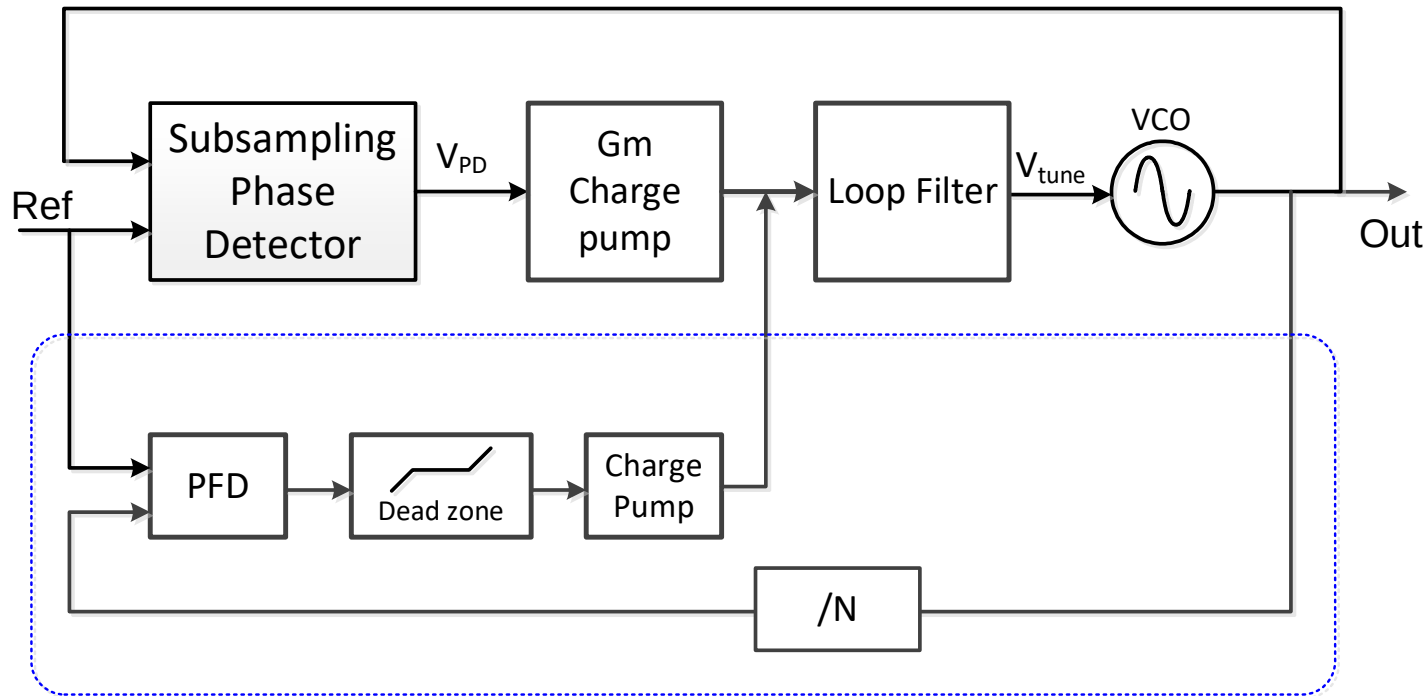
Subsampling PLL Model



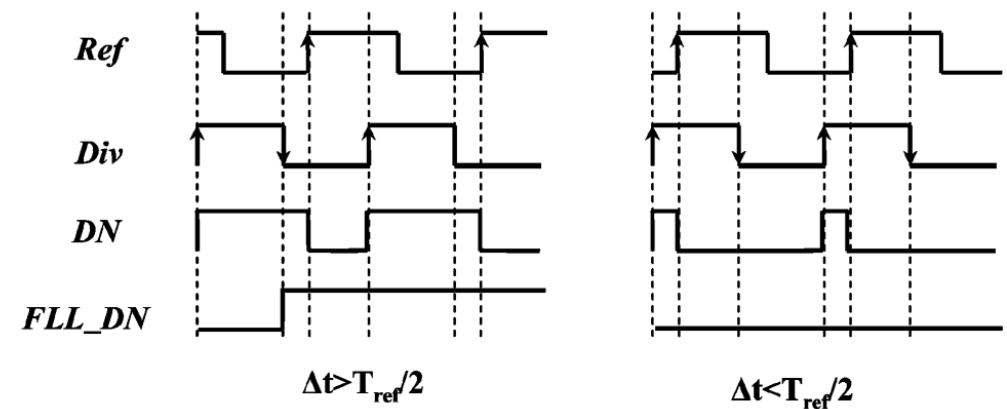
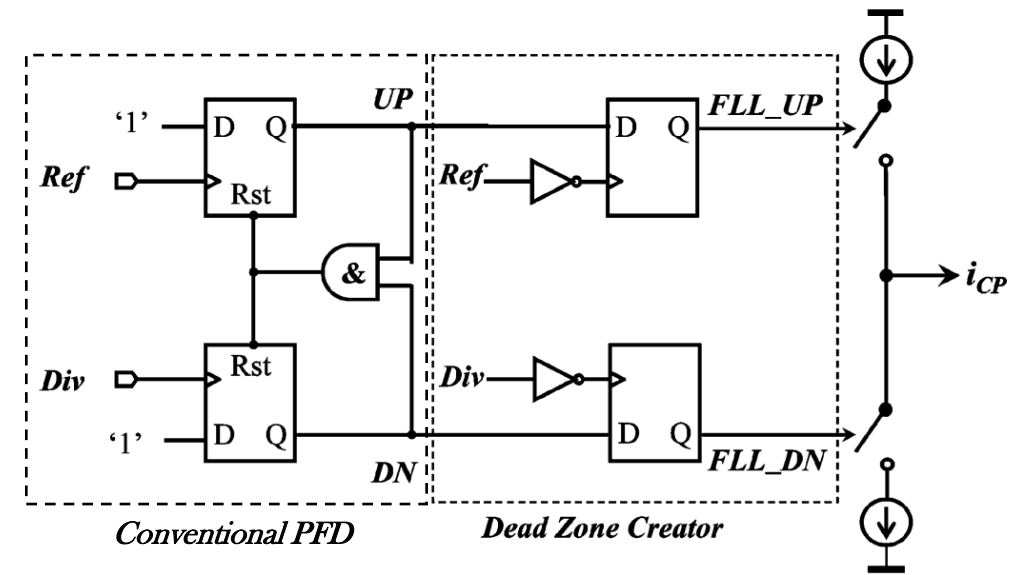
- Sampling PD gain: $K_{SPD} = \frac{dV}{dt} \frac{T_{ref}}{2\pi} = SR \frac{T_{ref}}{2\pi} = \frac{A_{VCO} \omega_{VCO} T_{ref}}{2\pi} = \frac{A_{VCO} f_{VCO}}{f_{ref}} = N \cdot A_{VCO}$
- Total SPD and Gm charge pump gain: $\beta_{CP,SS} = \frac{\Delta I_{CP}}{\Delta \Phi} = N \cdot A_{VCO} \cdot g_m \frac{\tau_P}{T_{ref}} = N \cdot A_{VCO} \frac{2I_{CP}}{V_{gs,eff}} \frac{\tau_P}{T_{ref}}$
- Compare with PFD-CP PLL: $\beta_{CP,PFD} = \frac{I_{CP}}{2\pi}$
 - $\frac{\beta_{CP,SS}}{\beta_{CP,PFD}} = 4\pi N \frac{A_{VCO}}{V_{gs,eff}} \frac{\tau_P}{T_{ref}}$
- Type-I PLL: SPD output voltage followed by a RC filter to control VCO
- Type-II PLL: SPD output voltage to control charge pump followed by RC filter



Subsampling PLL – Frequency Lock Loop

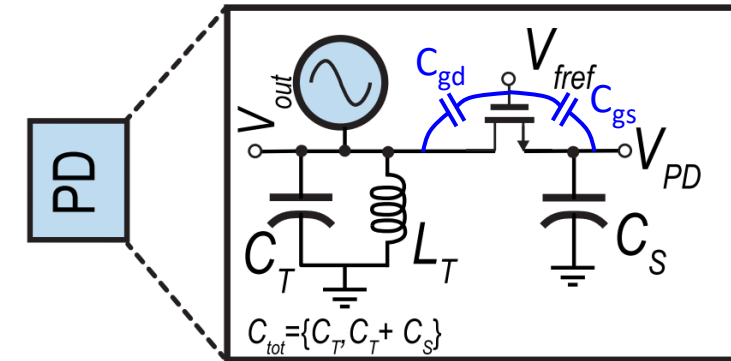


- PFD/CP + N-div as frequency lock loop (FLL)
 - Frequency detection ability
 - N-ambiguity issue solved
 - Fast locking
- Dead zone block is added to disengage FLL
 - Avoid FLL noise contribution



Subsampling PLL Reference Spur Issue

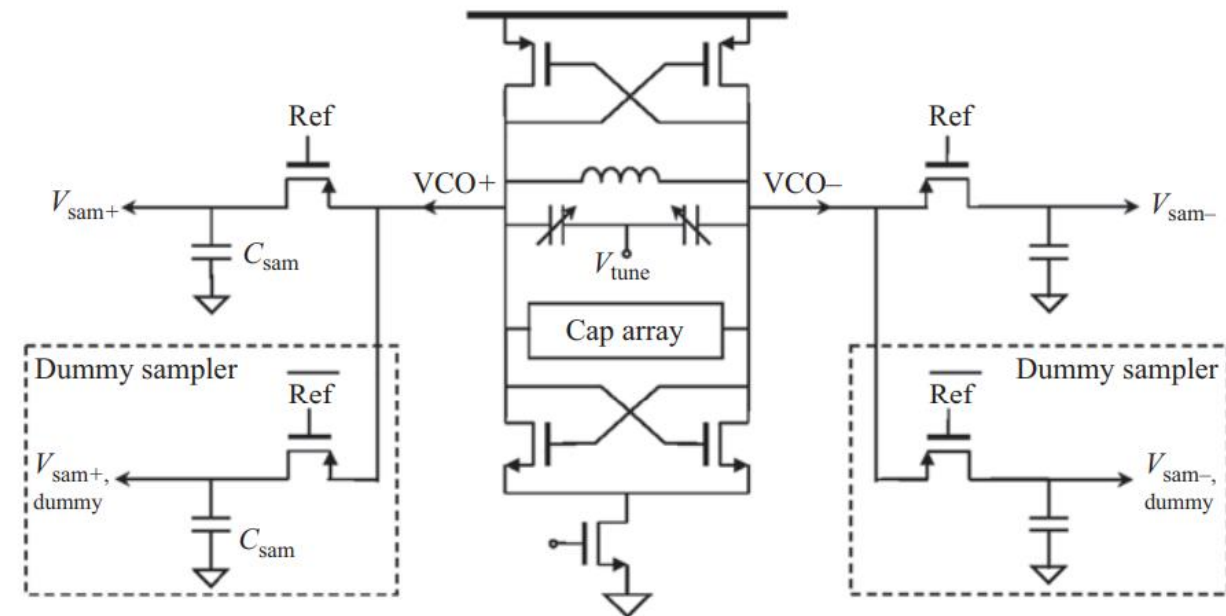
- Sampling PD switch and sampling cap C_S directly connected to VCO tank
 - Sampler loading capacitance is different between switch on and off state, toggling reference frequency – reference spurs (-40 ~ -50dBc range)
 - $C_{ON} = C_T + C_S + C_{gs,ON} + C_{gd,ON}$
 - $C_{OFF} = C_T + C_{gd,OFF}$
 - Clock feedthrough and charge injection



-40 ~ -50dBc

Solutions

- Add VCO buffer (~ -80dB range) – buffer output slew rate very high; extra power
- Dummy sampler clocked at the complementary of reference signal to compensate/cancel loading capacitance change – finite cancellation due to mismatch (-50 ~ -60dBc range)

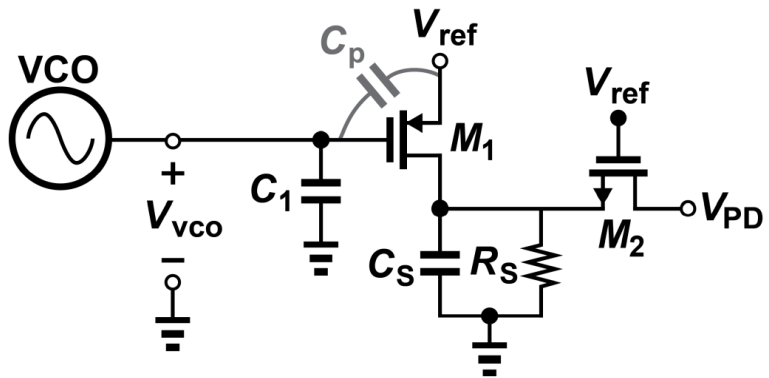


-50 ~ -60dBc

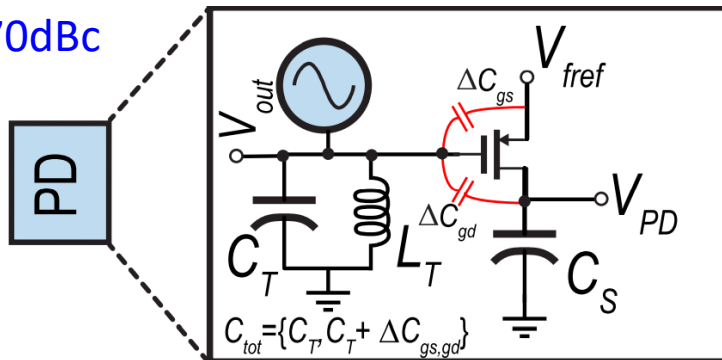
Ref [2]



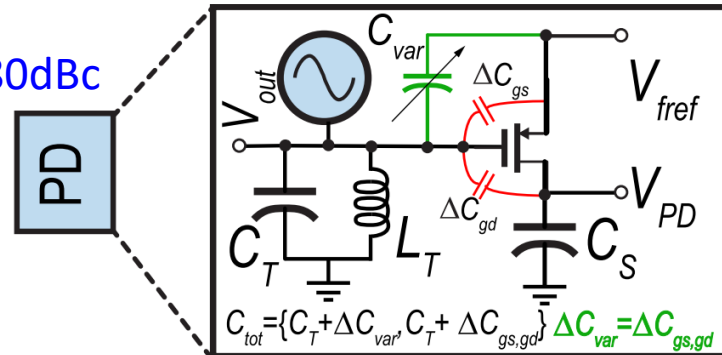
Subsampling PLL Reference Spur Issue



-60 ~ -70dBc



-70 ~ -80dBc

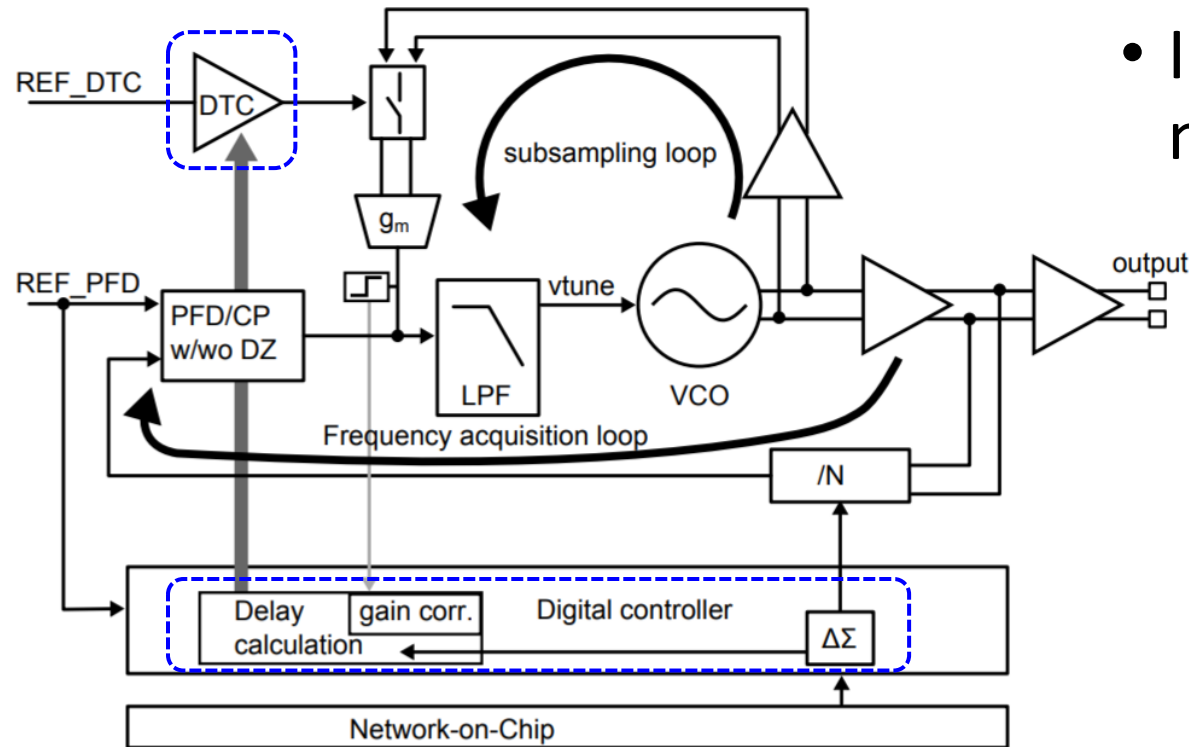


Ref [3,4]

- Switch VCO and V_{ref} ports
 - Sampling switch M1 becomes a single-ended single-balanced mixer – an active switch
- Much better isolation from gate to drain(mixer) than source to drain (switch)
- Capacitance in on and off states
 - $C_{ON} = C_T + C_{gs,ON} + \frac{C_S \cdot C_{gd,ON}}{C_S + C_{gd,ON}} \approx C_T + C_{gs,ON} + C_{gd,ON}$
 - $C_{OFF} = C_T + C_{gs,OFF} + \frac{C_S \cdot C_{gd,OFF}}{C_S + C_{gd,OFF}} \approx C_T + C_{gs,OFF} + C_{gd,OFF}$
- To further reduce $\Delta C = C_{ON} - C_{OFF}$, add C_{var} to compensate ΔC . C_{var} changes in the opposite direction as C_{ON} and C_{OFF} .



DTC in Sub-Sampling PLL

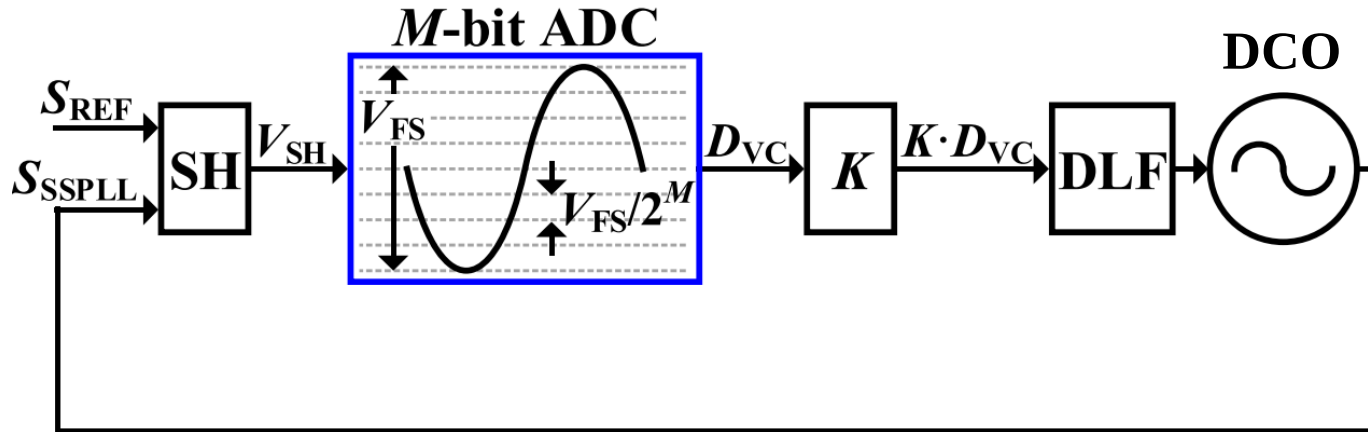


- In fractional-N PLL, VCO zero crossing is not always aligned with F_r .
 - F_v is not aligned with F_r at every edge
 - F_v and F_r rising edges may be off by +/- a few VCO cycles depending on DSM scheme
 - DTC moves F_r to align with F_v edge that aligns F_r to VCO zero crossing
 - DTC delay needed for alignment is determined by instantaneous N divide ratio (DSM output)

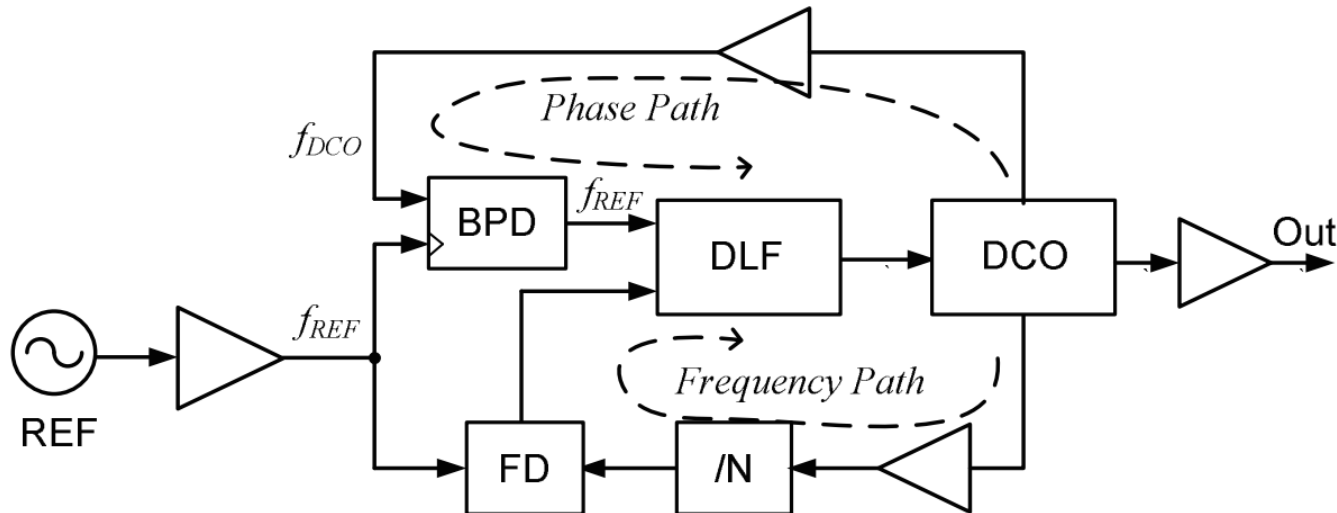
Ref [5]



Digital Subsampling PLL



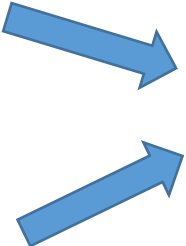
- Convert sampled voltage to digital with an ADC
- Followed by digital loop filter and DCO



- 1-bit digital sampler (BBPD)
- Followed by digital loop filter and DCO

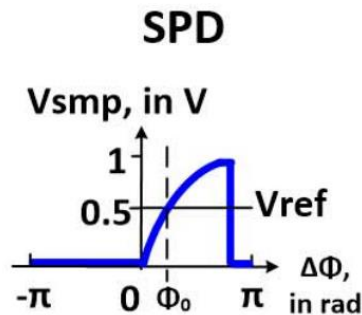
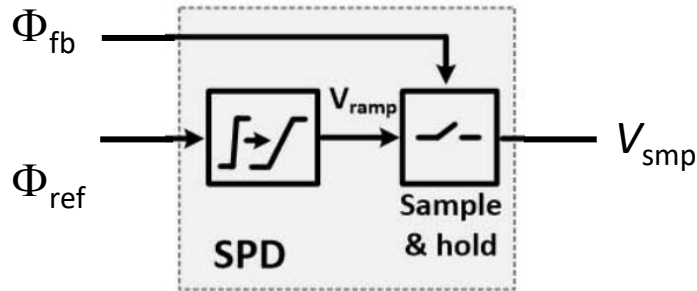


Issues in Subsampling PLL – Sampling PLL as a Solution

- N ambiguity, frequency lock loop (FLL) is needed
 - Reference spur, VCO tank is periodically disturbed by sampler loading
- 
- Include N-div in the feedback path
 - Sample N-div output, instead of VCO output
-
- Because of N divider, sampling PLL instead of subsampling PLL
 - Divider output is digital signal – need a slope generator to convert it to a ramp signal with timing information
 - Sampler PD gain depends on slew rate, lower than subsampling PD gain
 - Common: integer-N PLL, use DTC for fractional-N operation



Sampling PLL – Sampling Phase Detector



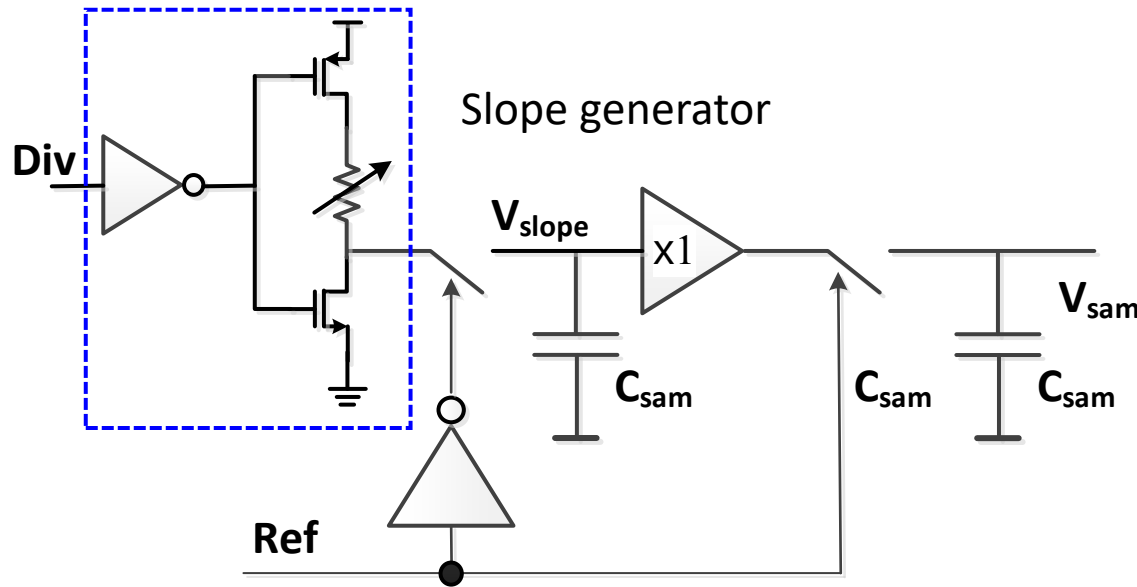
$$K_{SPD} = \frac{\Delta V}{\Delta \phi} = \frac{\Delta V}{\Delta T} \cdot \frac{\Delta T}{\Delta \phi} = K_{slope} \frac{T_{ref}}{2\pi} = \frac{K_{slope}}{2\pi f_{ref}}$$

Ref [6]

- Reference signal goes through a slope generator and becomes a ramping signal
- Feedback signal (VCO signal divided by N) samples the ramping reference
- The sampled voltage carries the timing or phase information.
- In the steady state after loop locks, the sampled voltage equals to V_{ref}
- $K_{SPD} = \Delta V / \Delta \phi = K_{slope} / (2\pi f_{ref})$
- High sampling phase detector gain for large K_{SPD} → suppress phase noise from following stages
- Integer-N PLL by nature. Use DTC to implement fractional-N functionality



Sampling Phase Detector Implementation



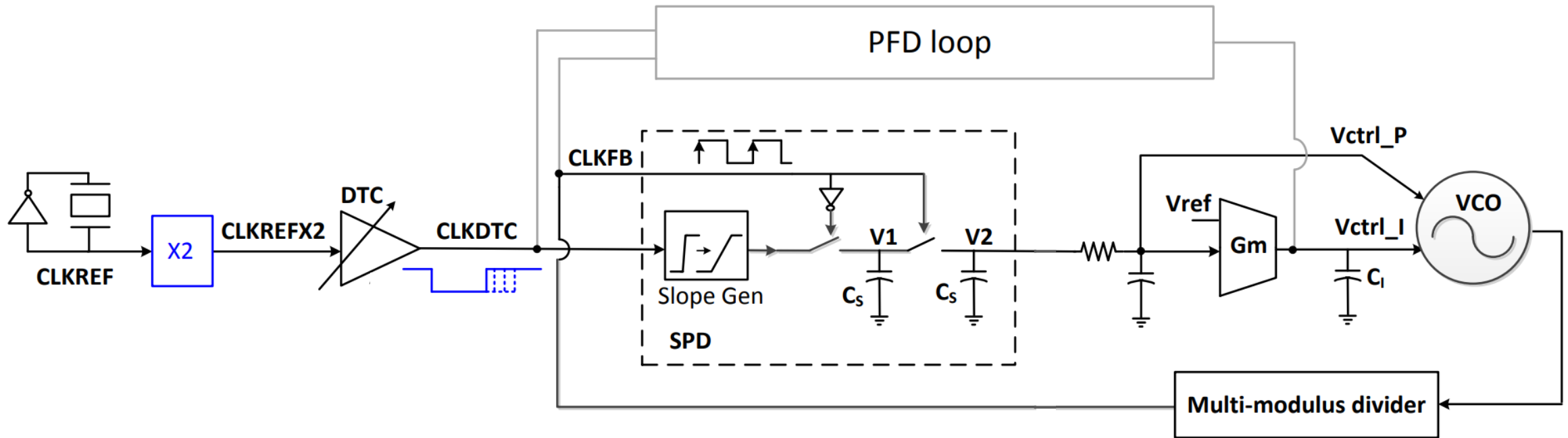
- $K_{SPD} = \Delta v / \Delta \varphi = K_{slope} / (2\pi f_{ref})$
- Parameter example:
 - $f_{ref} = 100\text{MHz}$
 - $K_{slope} = 5\text{Gv/s}$
 - $K_{SPD} = 50\text{v}/2\pi$

- Inverter with RC delay as slope generator
- Much higher phase detector gain than conventional PD or PFD
- Nonlinear RC charging – nonlinearity not an issue
 - Integer PLL: after locking, $V_{sam} = V_{ref}$, fixed V_{slope} operating point
 - Fractional PLL: DTC aligns Div or Ref edges to within a few pico-second, V_{sam} variation within a few millivolts.

Ref [6]



Fractional Sampling PLL with DTC

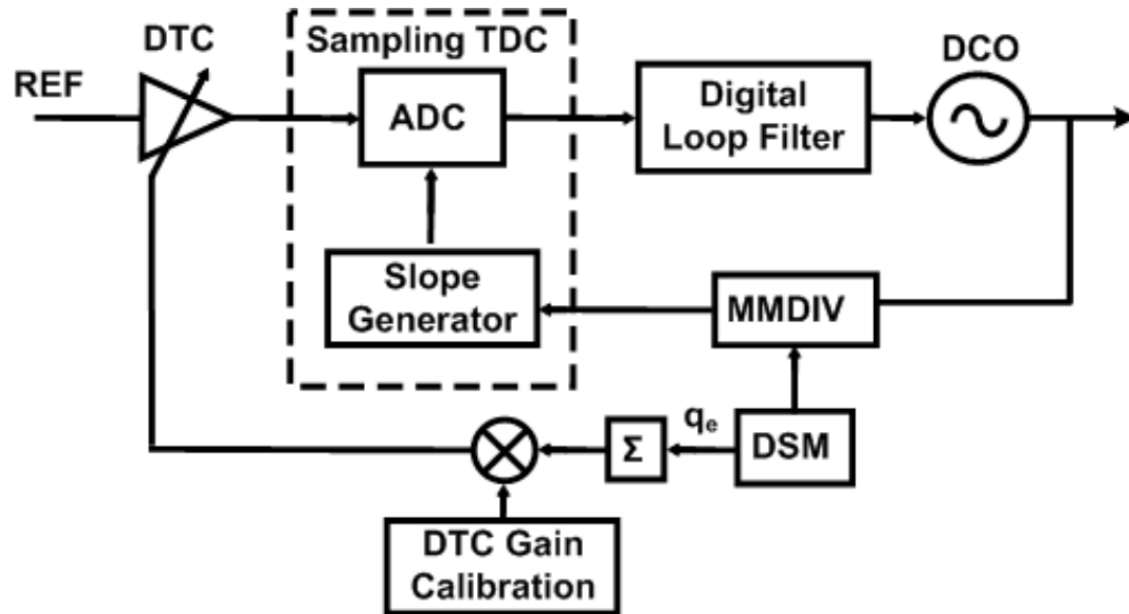


- Sampling PLL with N-divider in feedback path avoid N ambiguity
- PFD loop to speedup initial frequency locking. Powered down after lock
- Proportional path and integration path together form a type-II PLL (derive loop model and transfer function as an exercise)

Ref [6]



Digital Sampling PLL

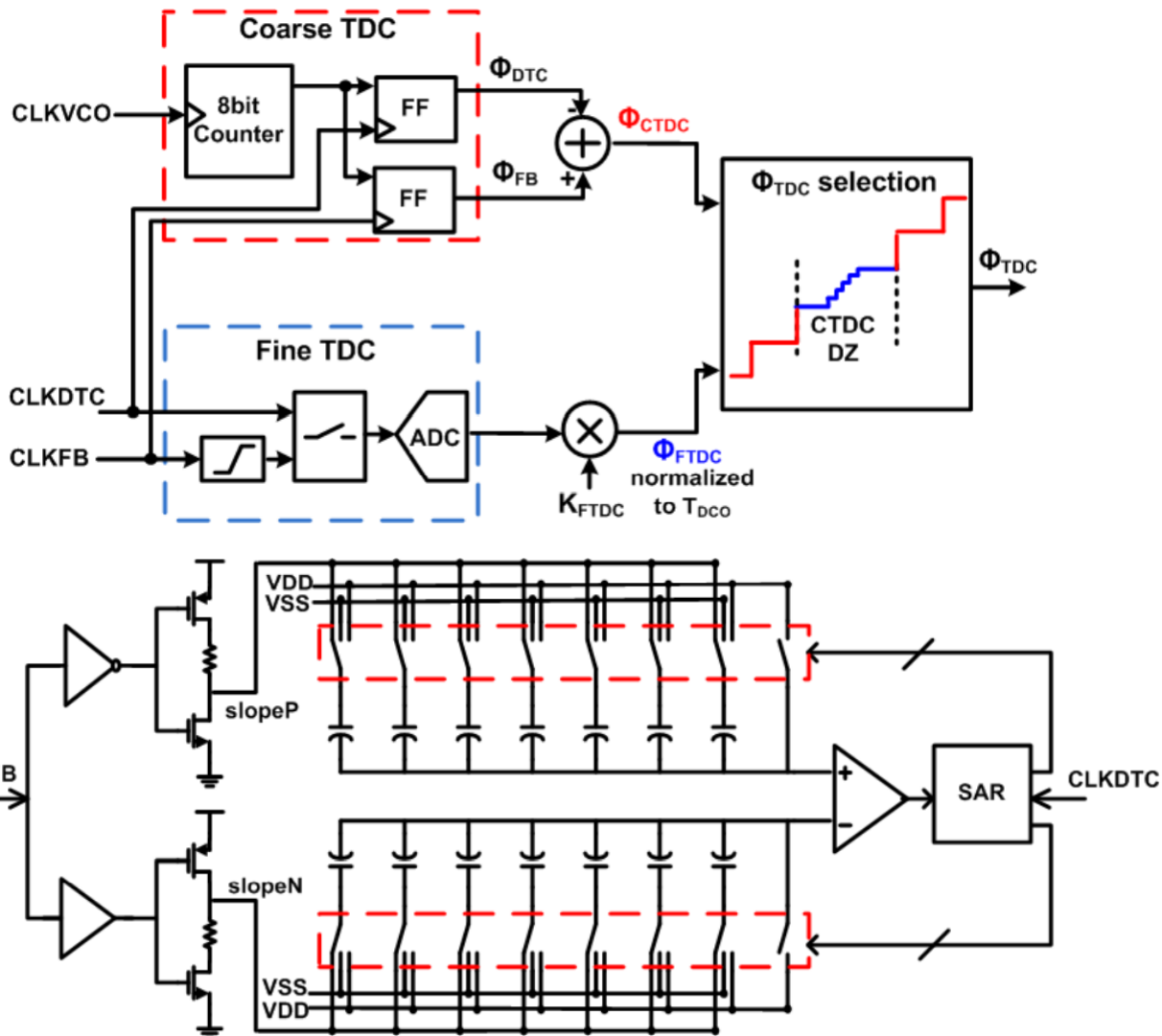


- Similarities with analog sampling PLL
 - N-divider in feedback path
 - Slope generator in sampler
 - DTC for fractional-N functionality
- Difference from analog SPLL
 - Sampling TDC instead of sampling phase detector
 - Digital loop filter instead of Gm charge pump and analog loop filter
 - DCO instead of VCO

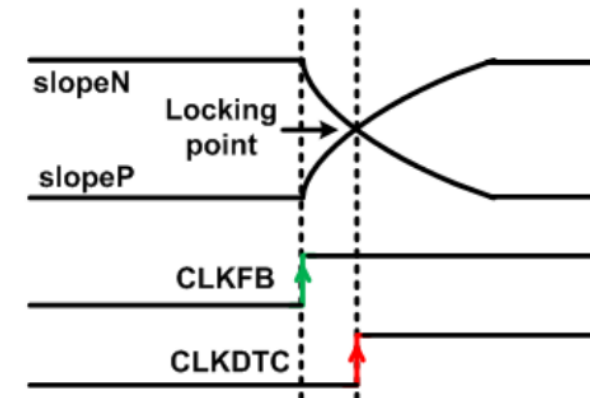
Ref [7]



Digital Sampling PLL – Sampling TDC



- Coarse TDC to handle large phase range – initial frequency locking
 - Implemented by counter, similar with VCO coarse tune engine
- Fine TDC for high resolution and low quantization noise
 - Implemented by SAR ADC
 - SAR ADC capacitor bank is reused as sampler capacitor
- Divider output (CLK_{FB}) as sloped signal
- Reference signal after DTC (CLK_{DTC}) as sampling clock



Comparison between Subsampling PLL and Sampling PLL

	Subsampling PLL	Sampling PLL
Feedback path	No divider	N-divider
Integer-N PLL	Yes Fractional-N with DTC	Yes Fractional-N with DTC
PD gain	Very high	High
Reference spur	Poor	Good
Auxiliary loop	PFD loop to speed up locking and avoid N ambiguity	PFD loop to speed up locking (optional)



References

- 1) X. Gao, E. Klumperink, M. Bohsali and B. Nauta, A low noise sub-sampling PLL in which divider noise is eliminated and PFD/CP noise is not multiplied by N^2 , JSSC 2009
- 2) W. Rhee, Phase-Locked Frequency Generation and Clocking, 2020
- 3) D. Lee, et. al., A Sub-mW 2.4-GHz Active-Mixer-Adopted Sub-Sampling PLL Achieving an FoM of -256 dB, JSSC 2020
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- 5) K. Raczkowski, , et. al., A 9.2-12.7GHz wideband fractional-N subsampling PLL in 28nm CMOS with 280fs RMS jitter, RFIC 2014
- 6) W. Wu, et. al., A 5.5-7.3 GHz Analog Fractional-N Sampling PLL in 28-nm CMOS with 75 fsrms Jitter and -249.7 dB FoM, RFIC 2019
- 7) X. Gao, A 2.7-to-4.3GHz, 0.16psrms-Jitter, -246.8 dB-FOM, Digital Fractional-N Sampling PLL in 28nm CMOS, ISSCC 2016

